

Your Sleep Blueprint

Evidence-based strategies personalized to your chronotype. Adapted from the Morningness-Eveningness Questionnaire (Horne & Ostberg 1976) and Munich Chronotype Questionnaire (Roenneberg et al. 2003).

1 What's Your Chronotype?

Q1

If free to plan your day, when would you wake up?

- A) Before 6:30 AM
- B) 6:30 - 7:30 AM
- C) 8:00 - 9:30 AM
- D) After 9:30 AM

Q2

When do you feel sharpest for mental work?

- A) Early morning (6 - 9 AM)
- B) Mid-morning (9 AM - 12 PM)
- C) Afternoon to evening (2 - 8 PM)
- D) Late night (after 9 PM)

Q3

How do you feel 30 minutes after waking?

- A) Wide awake, ready to go
- B) Alert after a few minutes
- C) Groggy, need 30+ min
- D) Foggy, sleep was light/broken

Q4

If you had to do intense exercise, when?

- A) 6 - 8 AM
- B) 8 - 11 AM or 3 - 5 PM
- C) 5 - 8 PM
- D) Whenever - hard to schedule

Q5

How would you describe your sleep quality?

- A) Fall asleep fast, wake early
- B) Generally good, normal
- C) Hard to sleep before midnight
- D) Light sleeper, rarely rested

Q6

On a free weekend, when do you fall asleep?

- A) Before 10 PM
- B) 10 PM - 11:30 PM
- C) Midnight - 1:30 AM
- D) Variable - no pattern

Mostly A

Morning Lark

Advanced circadian phase

Mostly B

Intermediate

Solar-aligned rhythm

Mostly C

Evening Owl

Delayed circadian phase

Mostly D

Sensitive Sleeper

Variable / high arousal

2 Your Chronotype Protocol



Morning Lark

~15-25% of population. Earlier dim-light melatonin onset (DLMO).

Sleep Window	21:30 - 05:30
Peak Focus	06:00 - 10:00
Caffeine Cutoff	11:00 AM

- Hardest tasks first - brain sharpest at dawn
- Exercise 6-7 AM to reinforce early phase
- Dim lights by 8 PM
- Don't sleep in >1h on weekends



Intermediate

~50-55% of population. Well-adapted to standard schedules.

Sleep Window	22:30 - 06:30
Peak Focus	09:00 - 13:00
Caffeine Cutoff	1:00 PM

- Morning sunlight within 30 min of waking
- Creative work before lunch, admin PM
- Exercise 8-11 AM or 3-5 PM
- Regularity is your superpower



Evening Owl



Sensitive Sleeper

~15-25% of population. Higher social jet lag risk.

Sleep Window	00:00 - 07:30
Peak Focus	17:00 - 21:00
Caffeine Cutoff	2:00 PM

- Bright light 15 min upon waking
- Melatonin 0.3-0.5mg, 5-7h before sleep
- Morning exercise to shift your clock
- Protect your late creative window

~10% of population. Often linked to hyperarousal or anxiety.

Sleep Window	23:30 - 06:30
Peak Focus	10:00 - 14:00
Caffeine Cutoff	11:00 AM

- CBT-I techniques are key for you
- No clock-watching, remove clocks
- Body scan or yoga nidra before sleep
- Consistent wake time is #1 priority

3 The #1 Sleep Metric You're Ignoring

Sleep regularity predicts mortality better than sleep duration. A 2024 UK Biobank study (n=60,977) found the most regular sleepers had **20-48% lower all-cause mortality** - independent of how many hours they slept. Consistent bedtime and wake time outperforms hitting 8 hours on random schedules. [1]

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ADULTS WITH INSOMNIA GLOBALLY

\$411B

ANNUAL COST IN THE US ALONE

48%LOWER MORTALITY WITH REGULAR
SLEEP

4 Universal Sleep Rules

☀ Morning Protocol

- **Sunlight within 30 min** of waking - 10+ min outdoor light sets your master clock
- **Delay caffeine 90-120 min** after waking to clear adenosine naturally
- **Same wake time daily** (+/- 30 min), weekends included - the single most impactful habit
- **Morning exercise** amplifies phase-setting and boosts deep sleep

🌙 Evening Protocol

- **Dim lights 2-3h before bed** - overhead lighting suppresses melatonin by 50%+
- **Screen curfew 1h before bed** or use red/warm filters
- **No food 3h before bed** - late eating raises core temp and fragments sleep
- **Warm bath/shower 1-2h before** - cools core via vasodilation

🚫 The Big Three to Avoid

- **Alcohol within 4h of bed:** Suppresses REM, fragments architecture. 2-3 drinks = losing ~1 sleep cycle
- **Caffeine after your cutoff:** Half-life 5-8h (CYP1A2 genotype). A 3 PM espresso is 50% active at 9 PM
- **Irregular schedule:** Weekend sleep-ins disrupt your clock like flying 2-3 time zones ("social jet lag")

5 Bedroom Environment



Temperature

Cool room, warm body before bed. Core temp drop triggers sleep.

65-68°F / 18-20°C



Darkness

Even dim light (5-10 lux) suppresses melatonin and disrupts glucose.

Blackout + Eye Mask



Air Quality

CO2 above 1,000 ppm cuts sleep efficiency by 1.3%. [7]

< 750 ppm CO2



Sound

Consistent background beats silence. White/pink noise or earplugs.

< 30 dB



Bed = Sleep Only

No screens, no work. If awake >20 min, get up and reset.

No Exceptions



Phone Out

Charge in another room. Use an analog alarm clock.

Another Room

6 Evidence-Based Supplement Stack

SUPPLEMENT	DOSE	TIMING	HOW IT WORKS	EVIDENCE
Magnesium L-Threonate	144mg elemental (~2g Magtein)	30-60 min before bed	Crosses BBB, enhances GABA. 2024 RCT: +36 min total sleep [2]	Strong
Glycine	3g	30-60 min before bed	Lowers core body temp via vasodilation, promotes slow-wave sleep [6]	Strong
L-Theanine	100-400mg	30-60 min before bed	Promotes alpha brain waves, reduces anxiety without sedation [8]	Strong
Apigenin	50mg	30-60 min before bed	Chamomile-derived, mild anxiolytic via GABA-A modulation [8]	Moderate
Ashwagandha KSM-66	600mg	With dinner	Cortisol -23%. Takes 6-8 weeks for full effect. 12-month safety data [5]	Moderate
Myo-Inositol	900mg	Before bed, 3-4x/wk	Serotonin receptor resensitization - targets mid-night awakenings [8]	Emerging
Tart Cherry	480ml juice or capsule	With dinner	Natural melatonin precursors + anti-inflammatory anthocyanins [9]	Moderate
Melatonin	0.3-0.5mg only	5-7h before target sleep	Chronobiotic phase-shifter, NOT a sedative. Short-term use only [10]	Strong

The Huberman Stack: Mg L-Threonate (145mg) + L-Theanine (200-400mg) + Apigenin (50mg), 30-60 min before bed. Add Myo-Inositol (900mg) 3-4x/week if you wake at 2-3 AM. Start with one, add every 3-5 days. [8]

7 Coping Strategies

Jet Lag Protocol

- **Eastbound:** Advance 1h/day for 3 days before. Morning light at destination
- **Westbound:** Evening light at destination. Easier - body delays naturally
- **Melatonin:** 0.5mg fast-release at target bedtime, max 5 days
- **Meals:** Eat on destination schedule immediately
- 2025 data: full normalization takes **7+ days** [11]

Shift Work Survival

- **Light:** Bright first half of shift, blue-blockers last 2h
- **Anchor sleep:** Keep 4h at same time daily, even days off
- **Blackout bedroom:** Non-negotiable for daytime sleep
- **Pre-shift nap:** 20-min nap cuts errors by 34%
- **Caffeine:** First half of shift only [12]

Creatine for Sleep Deprivation

- **2024 evidence:** Buffers cognitive decline during sleep loss [3]
- **Results:** 16-29% better processing speed during 21h deprivation
- **Dose:** 3-5g daily creatine monohydrate
- Does NOT replace sleep - reduces the cognitive penalty

Middle-of-Night Waking

- **Don't check the time** - triggers anxiety loops
- **NSDR / Yoga Nidra:** 10-20 min restores 60%+ of nap benefit
- **Myo-Inositol:** 900mg before bed, 3-4x/week
- **20-min rule:** If awake, get up. Read dim. Return sleepy

8 Your 7-Day Quick Start

- | | |
|--|---|
| Day 1 Set a fixed wake time - commit for 7 days, including the weekend | Day 5 Start one supplement - Mg L-Threonate is highest-impact start |
| Day 2 Add morning sunlight - 10 min within 30 min of waking | Day 6 Temperature audit - set thermostat to 65-68°F / 18-20°C |
| Day 3 Move your caffeine cutoff earlier by 2 hours | Day 7 Rate sleep quality 1-10 vs Day 1. Most see 2-3 point gain |
| Day 4 Screen curfew - dim lights and no screens 1h before bed | |

9 What to Measure

Free (No Wearable)

- **Sleep Regularity:** Same bed/wake +/- 30 min daily
- **Sleep Diary:** Bed time, wake time, awakenings. 2 weeks min
- **Subjective Quality:** Rate 1-10 each morning
- **Daytime Alertness:** Epworth Sleepiness Scale (free online)

With a Wearable

- **Deep Sleep %:** Target >15%. Physical restoration
- **REM %:** Target >20%. Emotional regulation + memory
- **HRV Trend:** Weekly trend matters more than daily
- **Resting HR:** Lower = better. Spikes flag issues

Ready to Go Deeper?

This cheat sheet covers the fundamentals. A personalized sleep assessment identifies *your* specific bottlenecks and builds a protocol around your biology.

zoe-sleep.com

Book a consultation with Dr. Kawalski

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REFERENCES

- | | |
|---|---|
| [1] Windred DP et al. Sleep regularity index as a predictor of mortality. <i>Sleep</i> . 2024;47(1):zsad253. | [6] Bannai M, Kawai N. Glycine improves sleep quality. <i>J Pharmacol Sci</i> . 2012;118(2):145-8. |
| [2] Ates M et al. Magnesium L-threonate and sleep quality: RCT. <i>Nutrients</i> . 2024;16:2669. | [7] Strom-Tejsten P et al. Bedroom CO2 and sleep. <i>Build Environ</i> . 2024;247:111059. |
| [3] Dworak M et al. Creatine and cognitive function during sleep deprivation. <i>Sci Rep</i> . 2024;14:4349. | [8] Huberman A. Sleep Toolkit. <i>Huberman Lab</i> . 2023-2025. |
| [4] Forbes SC et al. Creatine and cognition: meta-analysis. <i>Front Nutr</i> . 2024;11:1424972. | [9] Barforoush NZ et al. Tart cherry and sleep. <i>Food Sci Nutr</i> . 2025;13(4):e70923. |
| [5] Cheah KL et al. Ashwagandha and sleep: meta-analysis. <i>PLoS ONE</i> . 2021;16(9):e0257843. | [10] Auld F et al. Melatonin in sleep disorders. <i>Sleep Med Rev</i> . 2017;34:10-22. |
| | [11] Horne & Ostberg. MEQ. <i>Int J Chronobiol</i> . 1976. Roenneberg et al. MCTQ. <i>J Biol Rhythms</i> . 2003. |
| | [12] Sletten TL et al. Light in shift workers. <i>Sleep Med Rev</i> . 2024;78:101990. |

This guide is for educational purposes only and does not constitute medical advice. Always consult your physician before starting any supplement or changing your sleep routine.